Scalability & Future Plan

Date	03 July 2026
Team ID	SWTID-2026-9440
Team Size	4
Team Leader	Bodanapu Sai Krishna Reddy
Team Member 1	Anand Ch
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Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	1 Marks

Scalability & Future Plan;

This document outlines how the Curent project solution can be scaled to handle larger user bases, increased data loads, or extended featuœs in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

	Planned Feature	Target Timeline	Expected Impact
phase (Short-term: —6 mos)	Automated Model Retraining & Hyperparameter Tumng Pipeline	03 Jan 2027	Maintained model performance over time and automated continuous 0 'nliration,,
Phase 2 (Mid-tem -12mos)	Feature Enrichment (ill teraction and demographic features)	03 July 2027	Improved model accuracy by 5-8% on edge case predictions.

phase 3 (1.0 term: —l mos)	Explainable Al (XAD Dashboard foradministrators	03 Jan 2028	Increased transparency in prediction decisions and enabling faimess

Current System Limitations:

S.No	Limitation	Impact	priority to Address (High Medium / Low)
	Limited historical data range (e.g., only 2 years).	Reduced model training dataset diversity and potential for temporal bias.	Medium
2	High prediction latency during peak usage hours (single worker).	Degraded user experience and increased request timeouts on the Flask app.	High
3	Basic feature engineering (missing •debt features).	Sub-optimal model accuracy for certain demographic groups arid edge	Mediun

Scalability Plan:

S.Nu	Scalability Aspect	Current State	Proposed Upgrade / Solution
	User Load	Single-instance, synchronous Flask application.	Implement with multi-workers and autascaling using DockertRender horizontally scaled worker"
2	Data Storage	Single, non-replicated local MySQL instance	Migrate to a managed MySQL cluster (e.g.. on Render) with read-replicas for load balallc ing and high availability.
3	Performance	Model inference executes on the main application thread	Offload inference requests to a separate, asynchronous task qyeue (e.g., Celery/Redis) and optimize model with ONNX.
4	Security	Basic API key authentication with unencrypted Pll storage.	l:nplernent OAuth 2 with JWT, role-based access contr01, full database-level encryption for all sensitive data.

Future Roadmap:

Scalability & Future Plan

Scalability Area / Phase	Proposed Upgrade / Enhancement	Expected Performance Impact	Target Timeline
Phase 2	Migrating deployed model services to autoscaling cloud infrastmcture (e.g., ECS with Fargate) and database replication (e.g., MySQL read-replicas). Implement content delivery netivork network (CDN) for static assets.	Improved system responsiveness during peak usage periods. Higher concurrent user support. Reduced application latency. Enhanced data availability and fault tolerance.	QI 2027

Phase 3	Integrate continuous model monitoring and drift detection APIs. Explore integration with advanced anomaly detection models or third-party datasets for enhanced prediction features.	Proactive identification of model performance issues. Potentially higher prediction accuracy. Expanded system capabilities.	Q3 2027
Phase 4	Hardening the system security layer. Developing a secured mobile application interface. Expanding model applicability to new related product domains. Implement CI/CD for automated pipeline.	Reduced vulnerability to cyber-attacks. Increased market reach and accessibility. Higher overall system reliability. Enhanced business value and strategic alignment.	QI 2028